EARTH AND SPACE

Waste and Our World

Kinds of waste, compost, recycling and reusing, how waste harms nature, and the three Rs



recycling



reusable bag



glass bottle



seedling

In this pack

Learn it: Waste and our world ★☆☆
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This pack belongs to: _____ **Class:** _____

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LEARN IT · B3 · WASTE

Waste and our world

Name: Class: Date:

Learning goals

- Explain the differences between waste, compost, recyclable and reusable materials.
- Give examples of solid, liquid and gas waste from plants, animals and people.
- Explain how waste harms nature and plan ways to reduce it with the three Rs.

Every day, people throw things away: food scraps, plastic bags, paper, bottles. This is **waste**. If we do not deal with it well, it pollutes our rivers, land and air.

Kinds of waste

	Solid waste	Liquid waste	Gas waste
plants	fallen leaves	water released from leaves	oxygen (useful to us!)
animals	dung (poo)	urine	carbon dioxide breathed out
people's homes	food scraps, plastic, paper	dirty water from washing	smoke from cooking fires
factories and vehicles	scrap metal, packaging	dirty water, chemicals	smoke and exhaust fumes

Sort your waste

Type	What it means	Examples
compostable	rots and becomes rich soil (compost)	fruit and vegetable peels, leaves, egg shells
recyclable	can be made into new things	paper, glass bottles, metal cans, some plastic
reusable	can be used again	glass jars, cloth bags, old clothes
waste to throw away	cannot be reused, recycled or composted	used tissues, some wrappers



compost



recycle



reuse



reuse



bin

How waste harms nature

- Plastic in rivers and drains causes **floods** and harms **fish** and **turtles**.
- Dirty water from factories **pollutes rivers** like the Buriganga.
- Smoke from burning rubbish makes the **air** unhealthy.
- Rubbish heaps attract **rats** and **flies** that spread disease.

The three Rs

- **Reduce**: use less. Take a cloth bag to the market.
- **Reuse**: use things again. Use a jar as a pencil pot.
- **Recycle**: make old things into new things. Give paper, metal and plastic to the *bhangari* collector.

And **compost** your food scraps to make rich soil for plants.



recycle



reduce



reuse



compost helps plants

Key words

waste	things we throw away
compost	rotted plant waste that makes soil rich
recycle	make into something new
reuse	use again
reduce	use less
pollution	harmful things in air, water or land

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FACT CHECK · B3 · WASTE

Sort the waste

★☆☆
Grade 3

Name: Class: Date:

Learning goals

- Sort waste.
- Name the three Rs.

1 Sort the waste. [6]

SORT THESE

dead leaves **banana peel** **metal can** **cloth bag** **newspaper** **glass jar**

Compost	Recycle	Reuse

2 Solid, liquid or gas waste? [4]

Waste	Solid, liquid or gas?
smoke from a fire	
dirty washing water	
a plastic bottle	
carbon dioxide we breathe out	

3 Draw a line to match. [3]

Draw a line to match each pair, or write the letter.

reduce ●	● **A** use again
reuse ●	● **B** make into something new
recycle ●	● **C** use less

THINK IT THROUGH · B3 · WASTE

Less waste, cleaner world

★★☆
Grade 3

Name: Class: Date:

Learning goals

- Explain harm from waste.
- Plan to reduce it.

1 **How can plastic bags cause floods in a city?** [2]

2 **Write three things your family can do to make less waste.** [3]

3 **Why is compost good for a garden?** [2]

INVESTIGATE · B3 · WASTE

What rots in the soil?

★★☆
Grade 3

Name: Class: Date:

Learning goals

- Observe over time.
- Record results.

Class 3B buried four things in soil in a pot and dug them up after four weeks.

1 **Complete the table.** [4]

Item	After 4 weeks	Rots (compostable)?
banana peel	mostly gone, soft and black	
paper	soft, falling apart	
plastic bag	no change	
metal bottle cap	no change (a little rust)	

2 **What does this tell you about plastic?** [2]

.....

.....

3 **Which living things helped the banana peel rot?** [1]

.....

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REAL-WORLD CASE STUDY · B3 · WASTE

Turning rubbish into compost

★★★
Grade 3

Name: **Class:** **Date:**

Learning goals

- See how a community reduces waste.

Science | Citizenship | Maths

In Dhaka, a lot of household rubbish is food and vegetable waste. In some neighbourhoods, workers collect this waste from homes and take it to a **compost plant**.

They pile it up in boxes. Worms and microbes break it down. After about two months it becomes dark, crumbly **compost**, which is sold to farmers and gardeners. Less rubbish goes to the dump, and farmers get good soil.

Based on community composting projects in Dhaka (example numbers below).

1 **Why is food waste good for making compost?** [2]

2 **A family makes 2 kg of food waste every day. How much in a week?** [1]

3 **Make a poster for your school about the three Rs.** [3]

Three Rs poster

REVIEW AND CHALLENGE · B3 · WASTE

Show what you know

★★★
Grade 3

Name: **Class:** **Date:**

Learning goals

- Use waste words.

1 Find the words.

R	C	O	M	P	O	S	T	U
E	P	L	A	S	T	I	C	G
U	A	J	C	E	V	V	M	I
S	V	T	F	W	A	S	T	E
E	A	R	E	C	Y	C	L	E
E	K	H	L	R	I	D	M	I
R	E	D	U	C	E	A	D	O
C	S	U	U	E	O	A	H	L
H	R	T	M	P	L	A	V	C

compost **plastic** **recycle** **reduce** **reuse** **waste**

2 Which can be composted? [1]

◯ a plastic bag
◯ a mango peel
◯ a glass bottle

3 Taking a cloth bag to the shop is... [1]

◯ reducing
◯ recycling
◯ burning

4 Challenge. Design a bin system for your classroom. [3]

Classroom bins

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
sort compost, recycle and reuse	○	○	○
name solid, liquid and gas waste	○	○	○
explain how waste harms nature	○	○	○
use the three Rs	○	○	○
test what rots in soil	○	○	○